

6.1a Equation of a Line in Slope y- Intercept Form

Example 1: Complete the following chart

$$y = mx + b$$

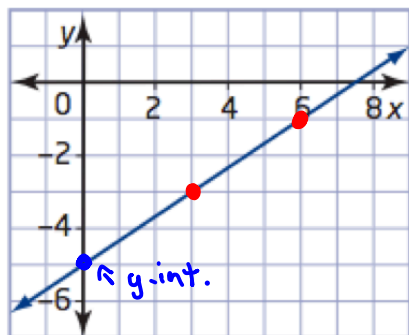
$\uparrow$ slope $\nwarrow$ y-int.

Equation	Slope	y-intercept
$y = -2x - 5$	$m = -2$	$b = -5$
$y = x + 2$	$m = 1$	$b = 2$
$y = \frac{2}{5}x + 8$	$m = \frac{2}{5}$	$b = 8$
$y = -\frac{1}{2}x$	$m = -\frac{1}{2}$	$b = 0$
$y = 4$	$m = 0$	$b = 4$

$\uparrow$
 $y = 0x + 4$

Example 2: Identify the slope and y-intercept of each line

a)



Slope: to find the slope use two points on the line

and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

Point 1: $(x_1, y_1) = (3, -3)$

Point 2: $(x_2, y_2) = (6, -1)$

$$m = \frac{-1 - (-3)}{6 - 3} = \frac{2}{3}$$

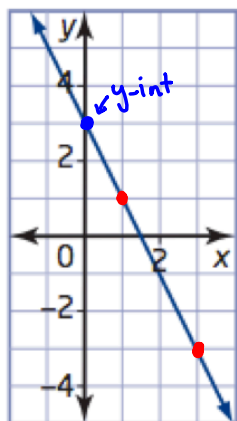
y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis. (When $x = 0$, $y = ?$)

Slope: $m = \frac{2}{3}$

y-intercept: $b = -5$

Equation of the line: $y = \frac{2}{3}x - 5$

b)



Slope: to find the slope use two points on the line

and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

Point 1: $(x_1, y_1) = (1, 1)$

Point 2: $(x_2, y_2) = (3, -3)$

$$m = \frac{-3 - 1}{3 - 1} = \frac{-4}{2} = -2$$

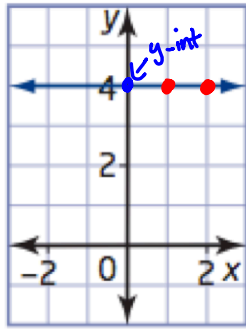
y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis. (When $x = 0$, $y = ?$)

Slope: $m = -2$

y-intercept: $b = 3$

Equation of the line: $y = -2x + 3$

c)



Slope: to find the slope use two points on the line and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

Point 1: (x_1, y_1) Point 2: (x_2, y_2)

$$m = \frac{4-4}{2-1}$$

$$= \frac{0}{1}$$

$$= 0$$

y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis. (When $x = 0$, $y = ?$)

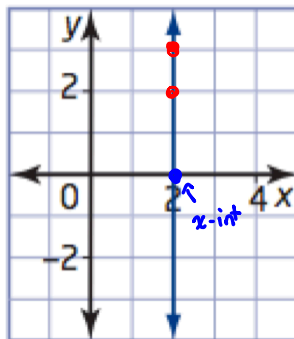
Slope: 0

y-intercept: 4

Equation of the line: $y = 0x + 4 \rightarrow y = 4$

Note: all horizontal lines have a slope of 0 and an equation of the form $y = b$, where b is the y-intercept.

d)



Slope: to find the slope use two points on the line and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

Point 1: (x_1, y_1) Point 2: (x_2, y_2)

$$m = \frac{3-2}{2-2}$$

$$= \frac{1}{0}$$

$$= \text{undefined}$$

y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis.

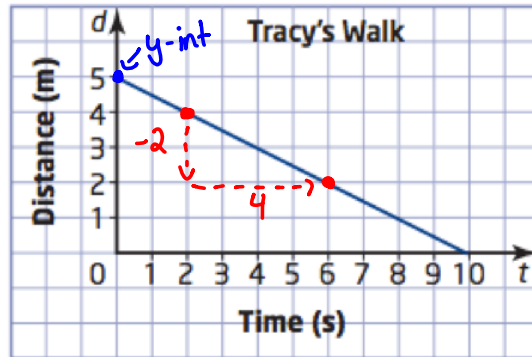
Slope: undefined

y-intercept: none

Equation of the line: $x = 2$

Note: All vertical lines have an undefined slope and an equation of the form $x = a$, where a is the x-intercept.

Example 3: Interpreting a Linear Relation



Identify the slope and the vertical intercept of the linear relation and explain what they mean.

Slope: $m = \frac{\text{rise}}{\text{run}}$

y-intercept: $b = 5$

$$m = -\frac{2}{4}$$

$$m = -\frac{1}{2} \text{ OR } -0.5$$

The slope represents Tracy's speed. The negative value means that her distance from the sensor is decreasing. Tracy's speed toward the sensor was 0.5 m/s.

y-intercept:

The y-intercept of 5 means that Tracy started walking at a distance of 5 meters from the sensor.

Equation of the relation:

$$y = -\frac{1}{2}x + 5$$